** Medium Level Questions (Real-Time Oriented)**

1. You are given a NumPy array of student marks. Write a program to extract marks greater than the class average.

```
import numpy as np
marks = np.array([70,80,50,20,10])
avg = np.mean(marks)
print("marks greater than the class average.",marks[marks>avg])
```

```
marks greater than the class average. [70 80 50]
```

2. A company stores daily sales data in a NumPy array. Write a program to find days where sales were higher than 50,000.

```
import numpy as np
sales = np.array([45000,50100,60700,49999,40000,65000])
print("days where sales were higher than 50,000.",sales[sales>50000])
```

```
days where sales were higher than 50,000. [50100 60700 65000]
```

3. Given a NumPy array of temperatures in Celsius, write a program to convert all values to Fahrenheit using vectorized operations.

```
import numpy as np
temp = np.array([30,40,34,32,38,-40])
faren = temp * 9/5 + 32
print("values to Fahrenheit",faren)
```

```
values to Fahrenheit [ 86.  104.   93.2  89.6 100.4 -40. ]
```

4. A NumPy array contains employee working hours for a week. Write a program to find employees who worked more than 40 hours.

```
import numpy as np
wh = np.array([40,39,42,41,38,42])
print("employees who worked more than 40 hours.",wh[wh>40])
```

```
employees who worked more than 40 hours. [42 41 42]
```

5. Given a NumPy array of product prices, write a program to apply a 10 percent discount on prices greater than 1,000.

```
import numpy as np
pp = np.array([1010,900,999,1000,1100,1200])
pp[pp>1000] * 0.9
print("10 percent discount on prices greater than 1,000.",pp)
```

```
10 percent discount on prices greater than 1,000. [1010  900  999 1000 1100 12
```

6. A NumPy array contains exam scores. Write a program to replace all scores below 35 with 35 using NumPy operations.

```
import numpy as np
marks = np.array([33,30,34,60,40,70,50])
marks[marks < 35 ] = 35
print("replaced all scores below 35 with 35",marks)
```

```
replaced all scores below 35 with 35 [35 35 35 60 40 70 50]
```

7. Given a 2D NumPy array representing marks of students in multiple subjects, write a program to calculate student-wise average marks.

```
import numpy as np
marks = np.array([[40,50,30],[60,40,30],[90,40,60]])
print("student-wise average marks.", np.average(marks, axis=1))
```

```
student-wise average marks. [40.         43.33333333 63.33333333]
```

8. A NumPy array contains monthly expenses. Write a program to identify expenses that are above the median value.

```
import numpy as np
expense = np.array([300,200,12000,1000,3000,5000])
med = np.median(expense)
print("expenses that are above the median value.", expense[expense>med])
```

```
expenses that are above the median value. [12000  3000  5000]
```

Hard Level Questions (Interview + Industry Thinking)

9. A 2D NumPy array represents sales data of products across different regions. Write a program to find the region with the highest total sales.

```
import numpy as np
sales = np.array([[50000,60000,40000],[45000,43000,30000],[70000,60000,80000]
region_sales = np.sum(sales,axis=0)
print("the region with the highest total sales.",np.max(region_sales))
```

```
the region with the highest total sales. 165000
```

10. Given a NumPy array of sensor readings, write a program to detect and extract all abnormal readings that are more than 2 times the mean value.

```
import numpy as np
readings = np.array([12,23,40,160,40,70,23,150,])
mv = np.mean(readings)
print("abnormal readings that are more than 2 times the mean value.", reading
```

```
abnormal readings that are more than 2 times the mean value. [160 150]
```

11. A NumPy array contains salary data of employees. Write a program to apply tax rules:

- Salary above 50,000 → deduct 10 percent
- Salary above 1,00,000 → deduct 20 percent

```
import numpy as np
salary = np.array([40000,52000,101000,130000,90000])
salary = np.where(salary>100000,salary*0.8,
         np.where(salary>50000,salary*0.9,salary))
print("Salaries after deducting taxes : ",salary)
```

```
Salaries after deducting taxes :  [ 40000.  46800.  80800. 104000.  81000.]
```

12. Given a 2D NumPy array of student marks, write a program to find how many students failed in each subject.

```
import numpy as np
marks = np.array([[40,50,30,50],[60,40,30,34],[90,40,60,31]])
count = np.sum(marks<35,axis=0)
print(count)
```

```
[0 0 2 2]
```

13. A NumPy array contains website response times in milliseconds. Write a program to count how many requests exceeded the acceptable threshold.

```
import numpy as np
rt = np.array([40,50,30,50,60,40,30,34,90,40,60,31])
tt = 40
count = np.sum(rt>tt)
print("Number of requests exceeded the acceptable threshold.",count)
```

```
Number of requests exceeded the acceptable threshold. 5
```

14. Given a NumPy array of daily profit and loss values, write a program to separate profit days and loss days using boolean indexing.

```
import numpy as np
Values = np.array([100,-60,110,-200,300,240,400,-230])
profit = Values[Values>0]
loss = Values[Values<0]
print(f"{np.sum(Values>0)} profit days.",profit)
print(f"{np.sum(Values<0)} loss days.",loss)
```

```
5 profit days. [100 110 300 240 400]
3 loss days. [ -60 -200 -230]
```

15. A 2D NumPy array represents attendance of employees where 1 means present and 0 means absent. Write a program to calculate attendance percentage of each employee.

```
import numpy as np
attendance = np.array([
    [1, 0, 1, 1, 1, 0, 1],
    [1, 1, 1, 1, 1, 1, 1],
    [0, 0, 0, 0, 0, 1, 1],
    [1, 0, 1, 0, 1, 0, 1]
])
days = attendance.shape[1]
present = np.sum(attendance, axis=1)
percentage = (present / days) * 100
for i in range(0,attendance.shape[0]):
  print(f"Attendance percentage of employee {i+1} is {percentage[i]:.2f}%")
```

```
Attendance percentage of employee 1 is 71.43%
Attendance percentage of employee 2 is 100.00%
Attendance percentage of employee 3 is 28.57%
Attendance percentage of employee 4 is 57.14%
```

16. A NumPy array contains ratings given by users. Write a program to normalize the ratings between 0 and 1.

```
import numpy as np
rating = np.array([1, 5, 8, 2, 10, 4, 7])
min = np.min(rating)
max = np.max(rating)
if max == min:
  print(np.zeros_like(rating))
else:
  normalize = (rating - min)/(max - min)
  print(normalize)
```

```
[0.         0.44444444 0.77777778 0.11111111 1.         0.33333333
 0.66666667]
```

17. Given a NumPy array of transaction amounts, write a program to flag suspicious transactions that are either extremely low or extremely high.

```
import numpy as np
transactions = np.array([40,5450,30,50,6550,4,30,34,950,40,3,51])
lt = 10
ht = 2000
suspicious = (transactions < lt) | (transactions > ht)
count = np.sum(suspicious)
print(f"Number of suspicious transactions: {count}")
```

```
Number of suspicious transactions: 4
```

18. A 2D NumPy array stores marks of students across tests. Write a program to identify the topper based on total marks.

```
import numpy as np
marks = np.array([[40,50,30],[60,40,30],[90,40,60]])
student = np.array(["Alice", "Bob", "Charlie"])
total = np.sum(marks, axis=1)
topper_index = np.argmax(total)
topper = student[topper_index]
print(f"The topper is: {topper}")
```

```
The topper is: Charlie
```